

Xiaoqi Wu

+86 13776020628 yukiwxq@gmail.com

Currently based in: London / Suzhou

Biological & Environmental Sciences Consulting | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Biodiversity and Conservation Biology, Global Change Biology, Ecology and Evolution, Behavioural Ecology, Field Skills in Ecology, Fundamentals of Ecology, Field Course in African Biology, Evolution and Diversity

Internship & Engagement

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Engineer · Analytical Department Shanghai 2024.07 - 2024.09

- **Decision-Ready Analysis:** Conducted stability testing of client-supplied drug samples using HPLC/UV, systematically identifying impurity risk through rigorous quality-control procedures to provide clients with analytical data usable directly for stability-assessment decisions
- **Process & Traceability:** Independently ran the full sample pre-treatment, instrument operation, and data-logging workflow, ensuring accuracy and traceability of results as a reliable basis for cross-departmental quality reporting

Shanghai Jianteng Education Technology Co., Ltd. Teaching Assistant · Biochemistry Department Shanghai 2023.07 - 2023.08

- **Teaching Delivery:** Provided extracurricular tutoring for students preparing for Oxford, Cambridge, and other top universities, independently delivering part of the course in English, mock interviews, and class discussions
- **Needs Analysis & Outcome:** Compiled and analyzed student needs documentation to give teachers differentiated instruction recommendations, helping multiple students secure Cambridge undergraduate offers – demonstrating structured analysis and client-facing communication

Research & Project Experience

Review: Drivers, Impacts, and Conservation Strategies for Insect Decline London 2024.03 - 2025.06

Literature review, critical analysis

- **Evidence Synthesis:** Systematically reviewed 70+ papers analyzing drivers (habitat loss, pesticide use, climate change), including a global meta-analysis reporting 2.5%/year insect biomass decline
- **Impact Assessment:** Assessed the impact of insect decline on ecosystem services (pollination, nutrient cycling), agriculture, and human health, producing a policy-relevant review
- **Critical Analysis:** Critically evaluated conflicting evidence, a "winners and losers" counter-narrative, and geographic/taxonomic reporting bias, proposing future research and policy directions

Drivers of Change in North Atlantic Fisheries Data and Food Web Construction London 2025.01 - 2025.02

Excel, literature review, food web analysis

- **Data Analysis:** Quantified a 90-year time series across 8 fish species, identifying species vulnerability such as sole's 40% decline (1990-2000) from bycatch and slow maturity
- **Recommendation:** Synthesized policy/climate/fishing-activity literature and proposed the common guillemot as an indicator species to inform fisheries resource management – report received high marks and distinction

Effect of Environmental Flower Colour Ratios on Insect Colour Preference Cape Town 2024.09 - 2024.10

Field experiment, R (t-test, Pearson correlation)

- **Experimental Design:** Designed a 5-colour pan-trap field experiment across 6 sites during a South Africa field course to study insect flower-visiting colour preferences relative to environmental flower colour ratios
- **Results:** Yellow, blue, and white traps all significantly attracted insects over baseline ($p < 0.001$); yellow capture rate was positively correlated with the proportion of yellow flowers in the environment ($r = 0.369$, $p = 0.027$), supporting the hypothesis that insects use common environmental flower colours as foraging cues

Skills & Other

- **Ecology & Fieldwork:** field experimental design (e.g. multi-site pan-trap colour-preference experiments), behavioural assays with inter-observer reliability testing, biodiversity/environmental surveying, stakeholder engagement and reporting, macroecology and phylogenetic comparative analysis (AVONET-scale trait data, PGLS)
- **Scientific Writing & Analysis:** literature synthesis and critical analysis, policy-relevant recommendation writing, R/Python statistical support (GLM, PGLS, Wilcoxon, t-tests)
- **Languages:** English (IELTS 7.5), Mandarin (native), French (A2) | **Interests:** piano, photography, skiing